

Algorithms, Correctness & Growth

A reusable Week 6 backdrop

COMSC-2043

Fall 2026 archive

What happens when the problem gets bigger?

- ▶ A few successful examples do not establish an efficiency claim.
- ▶ Count a meaningful operation, then ask how that count grows with n .
- ▶ Big O describes growth shape, not one machine's stopwatch.

Naive recursive Fibonacci repeats work

$$F(n) = F(n-1) + F(n-2),$$
$$F(5) = \underbrace{F(4)}_{\text{contains } F(3)} + \underbrace{F(3)}_{\text{repeats } F(3)}.$$

- ▶ The answer is correct.
- ▶ The same subproblems are recalculated again and again.

Correct does not mean efficient

- ▶ A contract tells us what the procedure must return.
- ▶ A trace and edge case test tell us more than friendly examples.
- ▶ Correctness and efficiency are different questions.

Same Fibonacci answer. Very different amount of work.

Bubble Sort: quadratic worst-case growth

$10n \Rightarrow$ roughly 100 times the n^2 work,

$$C(n) = \frac{n(n-1)}{2}.$$

- ▶ The 10/100/1,000/10,000/20,000/30,000 progression makes the cost visible.
- ▶ Hardware changes seconds; it does not change the quadratic shape.

Recursion was never the villain

Bubble Sort

- ▶ iterative
- ▶ $O(n^2)$ worst case

Merge Sort

- ▶ recursive
- ▶ about $O(n \log n)$

The growth pattern and repeated work matter more than whether a function calls itself.

Parallelism has a cover charge

- ▶ Divide work among workers.
- ▶ Start, coordinate, and synchronize them.
- ▶ Move data when the device is separate from the process.
- ▶ Small inputs can lose before parallel work has enough to do.

Measured CPU crossover

- ▶ The Fall 2026 reference uses 24 logical CPU workers.
- ▶ Parallel sorting first beats the sequential reference at $n = 10,000$.
- ▶ At large n , throughput separation becomes useful; at tiny n , clock noise and overhead dominate.

Measure the crossover; do not assume it.

GPU present versus process access

- ▶ Hardware can exist on the host.
- ▶ A sandbox or process may still be unable to see the device or CUDA toolchain.
- ▶ The Fall 2026 archive records the RTX 3060 Ti fact but no fabricated GPU timings.

Closing synthesis

- ▶ Algorithm choice changes growth.
- ▶ Parallelism changes throughput and constants.
- ▶ Correctness evidence, operation counts, and honest measurements belong together.
- ▶ Ask both questions: *Does it work?* and *How does the work grow?*